Pixels are fundamental units of display in various types of screens, including Cathode-Ray Tube (CRT), Light-Emitting Diode (LED), and Liquid Crystal Display (LCD) screens. However, the way pixels are defined and displayed can vary among these technologies:

- **CRT (Cathode-Ray Tube) Displays:**
- In a CRT display, pixels are defined as individual phosphorescent dots on the screen's surface.
- Each pixel corresponds to a tiny area on the screen's glass surface, which can emit light when struck by electrons from the electron gun at the back of the CRT.
- The color and intensity of each pixel can be controlled by varying the voltage applied to the electron gun and the type of phosphorescent material used.
- CRT displays typically use a pixel's physical position on the screen to determine its resolution, and the number of pixels is fixed based on the CRT's design.
- **LED (Light-Emitting Diode) Displays:**
- In LED displays, pixels are defined as individual LED elements arranged in a grid.
- Each pixel consists of one or more tiny LED elements (usually red, green, and blue) that emit colored light when electrically stimulated.
- The size and arrangement of these LED elements determine the pixel's size and resolution.
- LED displays can be found in various forms, such as LED monitors, LED TVs, and LED billboards.
- **LCD (Liquid Crystal Display) Screens:**
- In LCD screens, pixels are defined as individual cells containing liquid crystals.
- Each pixel consists of liquid crystal molecules that can change their orientation when subjected to an electric field. The orientation of liquid crystals determines the amount of light passing through.
- LCDs use a color filter array (e.g., RGB) in combination with a white backlight to create full-color pixels.
- Pixel size and resolution depend on factors like the size of the liquid crystal cells, the arrangement of pixels in a grid, and the screen's overall dimensions.